

ActInf Livestream #017.1 ~ Information flow in context-dependent hierarchical Bayesian inference

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hello and welcome everyone to active inference lab to the active inference live stream today is march 9th 2021 and we are in livestream 17.1 the active inference lab is an experiment in online team communication learning and practice related to active inference you can find us at the links and the contact mechanisms here this is recorded in an archived live stream so please provide us with feedback so that we can be improving on our work all backgrounds and perspectives are welcome here and we'll be using good video etiquette for live streams muting if we're not talking raising our hands so that we can hear from everyone on the stack and using respectful speech behavior here we are today in 17.1 coming fresh off of our fun quarterly roundtable number one and the first of two weeks discussing this paper with uh the author chris fields and we're gonna have a ton of time to get into it today and a follow-up discussion next week today our goal is to discuss and learn this awesome paper and area some areas that are probably new to everybody and today in 17.1 we're just gonna introduce ourselves say hello then chris is gonna give a presentation and then we'll be able to ask questions and continue the discussion from there so without further ado we can get into the introductions and warm-ups so everybody can just go around we'll introduce ourselves say hello and then we can also consider these warm-up questions so after introductions and anyone wanting to say something on a warm-up question we'll go to chris's presentation so i'm daniel i'm a post-doc in california and i'm excited today just to hear all of the things that we missed or messed in 17.0 and learn and uh recalibrate and correct and i'll pass it alex thanks hi everyone my name is alex watkin i'm in moscow russia i'm a researcher in systems management school and also co-organizer of active inference lab and they pass it to scott thank you alex my name is scott david i'm the director of the information risk research initiative at the university of washington applied physics lab i'm super excited about the the discussion today about information flows we deal in as name sounds and risk flows and information and risk are really hand in hand so this is going to be super exciting today and i'll pass it to siri uh hi my name is sarah davis i'm a

science artist and actually current grad student at um in philosophy of science at university of leipzig and i'm really looking forward to this talk today um who else do we have donna and chris and then so maybe shawna first and then chris you can introduce yourself and take it away hi my name is shanna and i'm a mathematician and um i'll be grad student in the fall but i've been doing all this stuff and chris and i are working on a couple cool things and i support him anyway hello cool so chris welcome thanks for joining us and we are looking at hello steven so chris um feel free to just introduce yourself and take it away with a presentation okay thanks daniel thanks for inviting me to suggest a paper and join you for this discussion session today i wanted to start with just a quick presentation to provide some background context on this paper which is about context uh written by myself and and jim glasbrook who is a mathematician uh we've been working together several years we actually connected and started working together because we were both interested in the perceptual and intentional phenotypes that one finds on the autism spectrum and that's where we encountered uh carl fristen actually and the whole active inference framework in some of his papers on autism so uh let me just start this little presentation and that'll give some background for here we go okay as i said just some some context for uh this paper about context and let's see oh okay so am i so good on on vision and hearing and all that looks perfect thank you okay cool so um here's a standard picture from carl friston on the theory of active inference uh it describes the interaction between some system that has internal states and some system that has external states and the interaction is mediated by sensory states and and active states of the form of arkhams blanket and when i first encountered this um it seemed like a very nice modern statement of ideas that i would trace back to robotics initially in terms of the the learn slash explore distinction that one finds in in developmental robotics especially and and even from there uh back to the basic theory in ethology where you have the approach avoid distinction um approach of being very similar to explore to acting on the world in order to learn something and avoid being a state in which the organism backs off and tries to figure out what it did wrong so the the theory has a very rich history and i think when it's what its current formulation uh due to fristen and his colleagues brings to it is a large body of applicable formalism and in fact i would characterize the theory of active inference as it's been presented as perception and cognition from a physics perspective and the presentation of the theory thus far has been primarily in the language of classical physics and i think that the language of classical physics is actually seriously unsuitable for talking about active inference it's complicated and messy and so what i'm going to do today is provide a little bit of background

from a from a more quantum theory perspective and i consider today just starting with a an introductory talk about quantum theory but i that won't give us quite the background that we need for this so um i won't be quite so uh straightforward about quantum theory but i am i'm happy to answer any questions about it and you can go on and on about it as shauna knows um so what i really want to to do is give you another way of looking at active inference which i would characterize as physics from the perspective of perception and cognition so taking the idea of observation and reformulating physics using that idea and you'll recognize from the paper there's a a a coco diagram that's sitting next to something that looks a lot like a markov blanket so the cocoon diagram represents the internal state of some system and the external state is completely unspecified all that's happening out there is some dynamics and the job of system a is to learn as much about system b as it can in order to make good predictions so this is still active inference just from a different point of view and looking at it this way it gets us into a a cloud of issues that are all related and i hope uh by this little introduction to show you how they're related at least somewhat give you a sense of how they're related and i'll i'll start this this cycle with system identification which is the old problem from engineering and cybernetics of figuring out what system you're working with and in physics this is typically just taken for granted and in biology it's often just taken for granted um but in uh computing theory for example and in what are called device independent protocols in physics now it's not taken for granted system identification is up for grabs intrinsic contextuality is the topic of the paper and it has to do with the unavoidable uncorrectable effects of unknown contextual variables on measurements being made now that's clearly related to the question of reference frames how am i making my measurements and in fact the first formulation of contextuality and physics by coca and specker was about making measurements with different kinds of apparatus and showing that experimental results could vary even for the same observable based on how what other things were measured at the same time this connects to the frame problem from ai which is the problem of of predicting side effects of your actions so side effects of making the measurement in a particular way would be an example this gets to the problem of memory if you're going to predict side effects you have to remember what you've done and so you have to ask about what are the side effects of recording your memory someplace because that's a physical action and finally we get to the question of ensemble sampling and the question of what probability actually means right prior probability is fundamental to bayesian decision theory but it becomes problematic to say what does prior probability actually mean and all of these issues swirl around at least in my view one issue

which is the issue of separability what does it mean to separate the world into a set of systems that then interact and separability has a very precise meaning in quantum so this is one reason for diving immediately into quantum theory in quantum theory the notion of separability is defined in terms of whether a quantum state factors into two quantum states so if i have a system and i can draw a boundary somewhere in it that divides it into two compartments a and b then if the joint state the state of the entire system a b factors into a state of a and a state of b then the system is separable and otherwise it's entangled and entanglement is kind of the natural state of systems in quantum theory whereas separability is the the only state of systems in classical so a has a state and b has a state and we can specify those two states independently of each other if this joint system is separate that means a and b have identities and we can talk about their interaction so i want to go back in history to the 18th century and and just mention how strange an idea this is if you think about physics at the time of laplace he was mixing the kind of atomism of democritus the kind of hard sphere irreducible particle atomism with the dynamics of newton and so laplace's world was a bunch of atoms each of which was a hard sphere a monad and they were moving around according to newton's laws and newton's laws act instantaneously so there's no sense in newton of forces being communicated the forces are just there and oh this is something that newton was aware of and here's a quote from nicholas guyson who's been very concerned about this issue of locality and physics and he in fact wrote a beautiful paper pointing out that the passion for locality in physics is very recent it comes from einstein einstein invented this idea that information travels at a finite speed before that everything happened instantaneously so newton understood that if you go up to the moon and move a rock then everybody's weight on earth changes instantly now that's contextuality that's a change here that happens unpreventably due to completely unknown events happening somewhere else so intrinsic contextuality is closely linked to non-locality now in laplace's world of all these atoms obeying newton's laws there aren't any boundaries there are no boundaries around systems like the earth or you or me uh or a cannonball so where do those boundaries come from in a laplacian world they have to be imposed from the outside so we impose a boundary we slice this picture in two and we say those are two different systems now if you think about that in classical that means that you've created an infinite potential that separates the particles from each other and prevents the newtonian dynamics from mixing them up and frank tipler several years ago wrote a beautiful paper published in pnas showing that if you just start with laplacian physics and you remove these singularities the boundaries around objects you get quantum theory automatically so laplace in a sense had a theory very

much like and he had a notion of the universe very much like entanglement who was full of what Einstein would later call spooky action in the distance okay so in this picture where we start with the system and we slice it down the middle and call the two parts a and b we're assuming that there's a boundary there it's not given to us by the physics and we're assuming that it stays constant over time and in quantum theory unlike classical the dynamics will automatically erase this boundary so quantum dynamics of an isolated system is unitary and so it mixes everything together into a superposition so if we say that a and b can be separated what we're effectively assuming is a weak local interaction that's only well defined for a short period of time because eventually the dynamics of the of the joint system will erase the boundary that we've put there okay but if we have a separable system and if it's finite dimensional we have a perfectly general way of talking about the interaction we can always write the interaction in this simple summation form uh the whole system interaction which i'll call h_{ab} with parentheses is the sum of what a is doing and what b is doing and then how they're interacting and this interaction can always be written in the form that i give that i've given below which is basically a bunch of thermodynamics numbers of degrees of freedom and kT and some measure of efficiency times a sum of operators and those operators we can choose a basis where those operators have binary outcomes so they question they correspond to asking yes no questions which is what John Wheeler postulated the very foundation of physics was just answers to yes no questions but we can always do this in quantum theory and that's why quantum theory is so nice it allows us to [Music] write down a formalism that explicitly describes the interaction between two systems in terms of exchanging information and this depends on an insight that Boltzmann had in the 1870s and it's really this insight that that gives us quantum theory Boltzmann realized that information always costs energy and what this did was rule out the way of thinking about measurement that one finds in classical physics which is that the observer is completely detached from the system and information flows from the system to the observer but but that process of observation doesn't affect the system and Boltzmann realized that this simply wasn't true observers aren't gods uh if i want to get information i need to spend some energy and that energy flows back into the world so by getting some information i'm always affecting what's going on so from a formal point of view these operators the m_k implement both perception and action and that becomes key of course to understanding active inference because active inference is about getting information and giving information by acting okay so to sum this up whenever i have a bipartite system that's finite and it's separable i can write the interaction as thermodynamics times questions and that

looks just like classical communication i can think of my two systems as agents alice and bob they're always called and they exchange bit strings finite bit strings so now let's think of a specific model let's suppose that alice and bob are each uh alternately preparing and then measuring the states of a bunch of independent cubits quantum bits and we can think of those as electron spins or something like that so um alice in some time uh prepares in electron spins and she hands those to bob and bob measures them and then bob writes a message by preparing the electron spins in some way and hands them to alice and alice measures them so they've now exchanged some bits but there's no requirement that they use the same z-axis to measure the spins so if they use exactly the same z-axis to measure their spins then if alice writes zero zero one bob is going to measure one zero zero one but if alice and bob are using different z-axes to measure their spins alice will write one zero zero one using her z axis and bob will get some probability distributions over 1 and 0 for each qubit measuring it with his axis and the way those probabilities will look will depend on how tilted his z-axis is with respect to analysis so this is now an interesting situation and this gets to daniel's question about shannon information versus other kinds information with semantics this is a purely quantum system there's nothing to introduce any classical noise so the information exchange is perfectly symmetrical and completely free of classical noise but by using different reference frames they introduce quantum noise and specifically what they lose is the message they lose the semantics from the message so alice encodes four bits bob gets four bits but the code may have been lost and the code resides in the z-axis if they're using the same z-axis they share a code if they're not they don't so semantics depends on shared reference frames and that's what shannon information theory is missing it's missing the idea of the reference frame that's used to decode the message it's just counting bits so shannon theory is just a quantitative theory of bits a way of counting bits and what we're really interested in is decoding encoding and decoding so we're really interested in reference frames and that will be a central theme it is a central theme in the paper okay so sum up every interaction between finite systems and a sufficient joint say can be represented in this way of two agents and they're separated by a set of bits a set of qubits that set of qubits can be thought of as a holographic screen it functions as a markov blanket and the information that he codes is completely specified it's the eigenvalues of the interaction written in its hamiltonian form so we know everything there is to know about an interaction in terms of a picture like this and these pictures are uh everywhere in biology um everyone is now using the market mark all blanket concept it's very common in computer science exactly the same concept is called an application programming interface and

it can be plugged down between two virtual machines and regulate their communication in cosmology the holographic principle was originally invented to deal with stretched horizons and black holes so it separates the exterior from the interior and it specifies exactly how much information can be obtained if you're an observer on the exterior about the interior and it's symmetrical so it also specifies exactly what the interior can learn about the exterior but you also see this concept in particle physics if you're thinking of a scattering interaction for example that's mediated by a vector boson of photon or something like that say it's electromagnetic scattering then all of the information is on that vector ozone there's no information about say entanglement on one side that's communicated to the other side it's it's just a single value a momentum transfer for example that's encoded by this holographic screen okay so where do we take this if we have an observer and she's got a holographic screen and she's interacting with the rest of the universe bob then by looking at her bit string alice can learn very little she doesn't know what z-axis bob is using so she gets a message but she doesn't know what code bob used she doesn't know how big bob's state space is so bob could be a small system could be an enormous system since she doesn't know the state space you can't possibly know bob's state she doesn't she doesn't know whether bob's state is separable or entangled so whether there are any divisions in bob that she has to worry about uh she doesn't know the dynamics on the other side so she can't predict what bob will do next and all of this is in principle uh looking out at the world i actually don't know that there are any separate objects in it i can't based on the information that i can get at my boundary which is just a set of interaction eigenvalues and so these questions have to be answered some other way they can't be answered by observation they have to be answered by a generative model so hence active inference is based on what models predict and what quantum theory tells us about generative models is that they're heuristic by definition they can't be constructed based on the data they're only heuristics so where did they come from and here when we look back at our picture our original picture of a joint system being divided uh recall that this boundary is arbitrary and it doesn't last very long so this joint state isn't separable for very long and the only asymptotic dynamics the only dynamics that's long lasting is the joint dynamics so alice's generative model which is implemented by her dynamics is effectively a sample of the joint it's just a temporary sample of what would be going on anyway if there was no division so it makes sense that alice's generative model contains information about the world but that information did not come from observation it came the time before the boundary was imposed and it will still be there after the boundary goes away okay another thing that quantum theory adds to the

acting inference picture is that it recognizes that all of the energy that's required to run the calculations also has to come through the markov blanket so if you think about the states on a holographic screen a markov blanket of any information encoding if that interface is the only interface between the two systems which is how we've defined it here then some fraction of those bits can't be informative because they have to be burned as fuel so by definition they're meaningless now this tells us again something very deep which is that the context of a measurement of can never be specified completely and that again is in principle because some of the information on the blanket has to be burned as fuel and we see this on the output side too of course of some of my output on the world some of my action on the world is just dissipating what for me is waste heat and i have no control over that that's part of my metabolism but i'm side affecting the world all the time by dissipating this waste heat which i don't think of as an action but which is an action from the point of view of the so here again we have an addition to the theory that we get from thinking about the dynamics from a quantum physics point of view instead of a classical physics point of view okay so this is the picture i showed before this set of qubits is the markov blanket of generated model is represented by this coconut diagram and it becomes clear of what the generative model is it's a combination of a reference frame and some combination some computation and it's clear what it does it specifies the option space for both perception and action it encodes everything that's detectable in everything that's actionable about b 4a so here just a couple of cartoons for example if alice looks out in the world and what she sees is a red rectangle interacting with a green diamond by some interaction then she must have reference frames for seeing rectangles and diamonds and she must have state observables for red and green and she must have a memory so that she can keep track of what's going on with those state observables long enough to construct this picture of an interaction so all of the semantics is inside the observer it has to be provided by these structures inside the observable observer so the semantics that's imposed on the world is imposed it comes from the generative model it's not out there in any sense part of of what the observer has to do is write memories if the observer is going to see change and since memories are classical data those memories are effectively written back in the blanket because the blanket is the only place where any classical data live so we can we can infer from this that if a memory looks internal to a system that system has to have intercompartmental boundaries inside it to to write those memories on so this actually tells us something gives us an expectation of physiology in organisms or in any sort of artificial structure that memories have to be written on boundaries of some sort so again here's this this

swirl of issues that all surround this question of separability of the world being separated into distinct pieces and i've mentioned all of them now except the frame problem which is the problem of predicting side effects and clearly if you don't have access to the dynamics outside you and you don't have access to all of your because some of them are just heat dissipation you can't possibly predict side effects all side effects so the frame problem is unsolvable in principle in this setting so in this paper what gemini set out to do was to construct a general representation of these reference frames and because they're physically implemented they're quantum reference frames they're not just abstracted and these things specify both what the agent can expect and what the agent could do so they completely define the option space for active inference for the agent so by having a general representation of the agent's reference frames we know in a sense everything interesting to know about the agent then we wanted to distinguish the reference frames that can be used simultaneously and these are the co-deployable reference frames from those that can't and the distinction again is given to us by quantum theory it's just operator commutativity so for example i can't deploy position and momentum at the same time because those operators don't commute they side affect each other it's the other way to think of it and physics is full of operators that don't commute so my my action space is is not unitary it's segmented into uh sets of things that i can't do at the same time without introducing contextuality so we showed that this non-co-deployability of reference frames actually induces intrinsic contextuality as it's defined in the literature what that means is that it creates undefined joint probability distributions um not just probability distributions that are difficult to make sense of or that needs some information added to them but probability distributions that actually can't be consistently defined why not because there's the information that would have to be specified to make them well behaved as information that you can't have that you can't get even in principle and then our final objective was to tie this explicitly to the frame problem which is a canonical problem that artificial intelligence discovered by mccarthy and hayes back in the unpredictability of side effects so that's that's it for my introduction kind of context setting and i am happy to um as soon as i can get to this button again switch to discussion chris thank you so much for that there it is yep we got it now thanks so much for that really amazing presentation i'm sure everybody was having some pretty interesting thoughts from what you were communicating really though thanks again for coming on the stream so um for those who are here please write down your questions and then when you ask your question let's stop after the question so that we can get a response to that question and not continue on with the question and then we'll raise our hands to hear from

everyone so i'm gonna first ask a question from the chat and then stephen i saw you raise your hand so anyone else raise your hand otherwise there's a couple questions from the chat so here's the first question from the chat can the context-based paradigm especially quantum contextuality help to answer how does spontaneously generated thought arise in the brain if yes how how do you see this relating to the brain um good question um i actually suspect that all cells are quantum computers and that to to really get a handle on information processing in cells we'll eventually have to use quantum information theory and the the reason for that is that cellular energy budgets are really small and and place a pretty strict upper a strict upper limit on how much classical information cells can encode and that upper limit is much much smaller than the information required to encode things like protein confirmation so i think that that cells are in fact uh devices where contextuality and entanglement play a role now what what spontaneous thought means i'm not sure oh i'm um very sympathetic to the idea that thoughts are and emotions and and everything else that we experience just the experience tip of a very large iceberg that we don't experience so i think our experiential access to what our minds are doing is extremely limited and um so how how what we actually experience as a as it were rises to the surface is uh a good question i tend to think that global workspace type theories are a good classical answer to that but all of those ideas i think will eventually need to be put in a much better mathematical framework thank you for the answer that's a long way of saying i don't know but it's what spontaneously came to mind so how could we ask for anything else stephen and then anyone else who wants to raise their hand yeah thank you um i was just curious um you mentioned sort of the quantum noise and the the the way that if you um have the z-axis um that you if you don't have the z-axis you can't decode back the information and often with quantum effects it's like it's when you have very low uh degrees of freedom for like a molecule or something that you particularly can see it compared to you know when it's in the melior of many degrees of freedom so i'm wondering if there's like a um certain scales where like the the interactions are in the cell or molecules are very quantum like and then it becomes more contextual and a bit more blurry and then another scale it becomes a bit more quantum like maybe once you start to get to brainwaves interacting and is the contextuality piece moving it in a way it's another way of moving it in and out of a more classical um kind of noisy space um so that it's like where it's got more um noise and it's less easy to see sort of more distinct quantum effects so i was wondering like is there particular scales at which you see the quantum effect being more like you think a traditional quantum like flipping between states um and if there's other scales where it's much more like fristen

talks about this kind of non-linear miller great yeah that's that's a very good question is the question about is there a quantum to classical transition and if so where that that question is typically posed in terms of decoherence and in in biological systems um estimates of decoherence times are anywhere from femtoseconds down there's the famous paper by tegmark from uh criticizing the hammer off penrose theory where um his argument against hammer off and penrose is basically the d coherence happen many orders of magnitude faster than than any cognitive processes or any neuronal processes so based on that type of reasoning biochemistry is thought of as being classical somewhere above the say finto second type of time scale so molecular dynamics calculations are basically classical when they're in like finto second types of time steps and that introduces the question of what is noise right what what does it mean to say that there's thermal noise in a system um when we think about thermal noise classically for example we might think about a protein that's sitting in a bath of water and the water molecules are thermally excited so they're banging into the protein like little billiard balls and that's in fact one way of thinking about a decoherence model so what we've introduced here is this idea of a temporal sample that we're looking at the protein molecule for a long enough period that the water molecules have had a chance to bang into it from many different directions and we're somehow averaging over that temporal sample so that gets to this question of ergodicity that came up in the earlier and it it makes us ask how am i implementing this this temporal sample and the the obvious answer is i'm implementing this temporal temporal sample by measuring the state of the protein in a particular way so i'm i'm using an enzyme say to to measure the state of the protein and the the enzymatic reaction has some particular time constant so it takes a certain amount of time for the enzymatic reaction to take place and during that time the protein may be wiggling around so i think that that all of these questions about classicality in the end come down to questions about measurement and the temporal resolution of measurements implemented by something an instrument or another part of the biological system or an observer of some kind so this is a this is kind of a long way of saying that i suspect that this question of classicality is always a question of observation and the the sort of coin in picture that that i outlined implies that the classical information only exists on the mark on blanket that there's no classical information anywhere else in the system so classical information may not in a sense be ontological at all there may be no ontological classicality uh if this kind of communication picture is correct now if that's the case noise is um epistemological noise is an artifact of making the observations in a certain way with a certain reference frame right an instrument is just a reference frame thank you for the deep

answer chris we're gonna go blue scott and then a question from the chat hi chris thanks for the um contextualization talk that was really nice um so i just was wondering you just mentioned earlier like the prospect of each cell being a quantum computer and i'm also like i have a similar kind of thought or theory and i just was wondering if you've heard of like quantum cognition and the positive molecule with the idea of these entangled pyrophosphates as as holding kind of quantum cognition cellularly have you heard this theory and do you think it's credible or do you have other suppositions as to where this quantum cognition could occur intracellularly i've certainly heard and seen various studies of localized quantum effects in particular molecular structures so stacked pi domains and and so on and there's the whole story about uh is there quantum coherence in um um goodness i've just dropped the word light harvesting molecules or electron transport or something like and um you know there's the hemorrhage theory that there's that there's quantum processing in microtubules and in all of these approaches one is thinking about a quantum process happening in a background of a classical cell once one's thinking the rest of the of the biochemical situation as being classical a classical environment of some kind in which this single quantum process is happening and as i sort of expressed in in answering the last question i i suspect that that entire way of thinking is probably wrong that we we can't actually think of bulk biochemistry as being classical so um trying to think of a particular event as being a quantum event against a classical background may just be misleading us about what's how information processing is actually occurring and if we think of of the interior of a cell say the entire interior of something like a bacterial cell as being in one coherent state then um it's not that that their particular quantum molecular states is it's that the entire interior is one big entangled mess that demands a very different way of thinking about biochemistry and i don't even know how to formulate that way of thinking about biochemistry but i suspect that approaching it as a as a quantum information problem where we're extremely explicit about where measurements are made would be what would have to be done thanks chris really amazing about the relational and the semantic biochemistry that's not how we got it in 103. so scott and then a question from the chris thank you that's a absolutely fascinating presentation a number of questions but i'll just ask one or two um one of the challenges of quantum or challenges to quantum physics being applied in the macro level is that you know don't take people have said don't take the concepts and apply them to social constructions and things but here the um origin of the active inference notions or the relationship to the acupuncturist suggests to me um the opportunity to look at phenomenon of much larger systems larger scale social scale um and so i wonder about

your um uh acceptance or enthusiasm for that notion of applying these ideas to uh social systems political systems economic systems other systems that have interactions that can be described with active inference um and this feels to me like a nice way to say hey we're talking about quantum politics markets we're talking about uh prionic governance with conformational changes of narratives um and so a lot of the work that we've been doing on the information side is gonna is i've been saying without knowing you've explained to me what i've been trying to say for years now in this talk and i've been saying without knowing why that data plus meaning equals information and the meaning part is what you just explained to me what i meant by meaning and i always thought of it i was a lawyer for 30 years so i always thought of it as the meaning like a contract with an enforceable narrative which is meaning but you've just given me a new way of looking at that meaning from a quantum perspective which is extremely robust and scale independent i just wonder about your comments on that and your enthusiasm for that kind of application thanks oh thank you that's that's a very interesting observation i think the experiments are going to be very difficult but i think if we can figure out how to do the experiments we're going to find contextuality and entanglement kinds of effects all over the place and i would i would point you to the work of of uh atamar zad zafarov and his colleagues some of which is referenced in the paper in in their formulation called contextuality by default what they've done is translate this kind of most standard model of quantum contextuality into classical probability theory and and given this criterion of if a classical probability a consistent classical probability distribution cannot be defined over a set of data then it displays contextuality and they they published a series of papers in the past four or five years on this approach and one of them deconstructs a large number of claims of quantum effects in psychological data which have been reported now by numerous groups around the world with various kinds of experimental protocols and there's active controversy between members of these different groups about what is and is not evidence of quantum so it's an enormous literature now but using the contextuality by default approach zafrov and colleagues his student cervantes is one of the lead people who's done this work have constructed extremely careful experiments where under their their best criteria there are robust intrinsic contextuality and interestingly they have to do with meaning so one that we refer to in the uh in the paper is called the snow queen experiment and it's based on uh interpretations of the hans christian andersen story so i think that as the tools and techniques for developing these these kinds of experimental protocols are improved uh we're going to see more and more uh effects that are very very difficult to challenge from a statistical point of view i mean the

problem is as therefore often and i'll point out is that in physics if i'm doing something like a bell experiment i can put my two observers far enough apart that they can't communicate just due to the speed of light in the time it takes to do the experiment and you can't do that in biological so you always have this communication loophole to deal with and what what zafrovan and colleagues are really trying to do is develop the statistical analysis techniques uh to get around the communication loophole in an arbitrary experiment you know carried out with college undergraduates and i think they've been successful thanks for the response so we're gonna go to a question in the chat and then if anyone here who hasn't raised their hand yet raises their hand otherwise we'll go back to um someone who has so this is from sudakar in the chat how will an agent acquire information for their you've uh said that there has to be some heuristic to infer the information even from the joint world and you said that the object semantics have to be inside the observer but how does the observer learn that in the first place for their that's that's an excellent question and again i the only answer i can give is i don't know i'll but i think that i think that biology gives us hints if you look at the process of cell which is one of the best examples of of separation that we have to deal with that we can really manipulate in the laboratory then and if you ask how much information is is shared between the daughter cells well clearly their genomes are shared but the genome is only a very tiny part of the shared information they also share information that's encoded in their cytoskeletons they share information that's encoded in the cell membrane the cytoplasm itself they may share aspects of their bioelectric states so there's a lot of information that is inherited if you will across cell division and most of it has not been quantitated in in any reasonable way i mean it's easy to quantitate the information in dna or protein but it's extremely difficult to quantitate this other information and i'll just give you one example of that's from my colleague mike levin's lab at tufts they work on regeneration and planaria and they've been able to show um experimentally that it's been known it's been known for quite some time that if you alter the ability of cells to communicate bioelectrically then you can get planaria to regenerate with two heads instead of one but what mike's lab was able to show is that this is epigenetically heritable so planaria that looked perfectly normal may have inherited the tendency to regenerate with two heads after injury and this happens without changing dna without changing rna without changing protein content without changing any of the kind of obviously inheritable factors and have been able to show experimentally that the bioelectric effect happens first before the observable changes in gene expression so here's a case where a completely uh unlooked form kind of information is epigenetically inheritable and makes a

huge difference to what the animal does with his with his physiology with its anatomy i mean the the two-headed planaria have two perfectly good operational brains that are independently able to direct behavior uh so they're they're pretty weird creatures thanks for the response and just briefly that's one reason why the information is in the genome kind of zoolander take doesn't work well because that's assuming there's this map between the source code in the program but then there's these alternate stable states that exist because the semantic or the epigenetic or the environmental frame so it's not quite as simple as just this one mapping that you made so that's a really nice point all right we're gonna have sarah and then shauna and then a question from chat thank you go ahead sarah yeah i um i have almost like two different questions or directions and i'm not sure how they're gonna match so i'll just start with one and see where it goes um one it seems to me like the the the way you frame the quantum um set up you know you have a you have b you could actually kind of do this with any problem it's probabilistic type notation that you're using but you know you have a you have b they have the joint distribution of the two and um it seems like i i guess the question that comes to my mind is like what happens when we don't if if we were hypothetically because i know our world is constructed around reductionism but if we were hypothetically able to separate out or to not think of things as um reductionist you know and so like for um how would our how could our math possibly be different if we were looking at things as only spanning sets rather than eigenvalues and eigenvectors um yeah so this is kind of where my mind goes is like is there a is there even a kind of a way to approach a problem purely through contextuality rather than than you know reductionism of any kind that's yeah hmm i don't know on the on the math front i may actually defer to shauna here as she she does know much more category theory than i do and i suspect the answer is a category theoretic type of but i'll make one comment which is that reductionism is one of these complicated words that has many different meanings for many different people but one way to think about what we're doing in quantum theory is merely labeling and we're labeling things for convenience with the full understanding that our labels in a sense don't have any physical they they only have meaning in the context of particular observational outcomes that we've made by measuring things in a certain way but that in itself is a problematic statement right and this is this is in a sense the core of the copenhagen interpretation it's it's what bohr talked about in his paper in nature in 1928. uh he basically said at the end of the day we have to talk about apparatus and we have to talk about colleagues and we have to talk about and in a sense quantum theory tells us all of that is um a poor approximation but it's the only language that we've got so uh i i think this this question of

of how far can we go without uh reifying separation is a very good question and as usual my answer is i don't know awesome thanks for the response you have any thoughts yeah let's go oh yeah i wonder if i should um ask my two yeah category theory is really powerful and when chris says there's only one logic no no no he and i are trying to find another one something called like a chromatic types and i was actually going to ask you chris if you think you know the phenomena of regeneration i think could actually be modeled by when we're trying to do that chromatic filtration right it seems like regeneration seems to be some kind of um morphism equivalent of when the stop mechanism is what does it mean to stop maybe that means stop means not making more morphisms and the two had a panera like actually has a two category structure right that has a one morphism and a two morphism and i think that's what's controlling the central nervous system right that it if you can have two morphisms and you could have this double head kind of thing and so whatever the stop mechanism is i think it's almost like remember when i was showing you those filtrations how you could actually make more equivalences and more equivalences using that boost field localization so i think you know there's another one called the topological localization where the objects and the categories are reflections and that might be what might help um sar a little bit is that there's different categorical models for interaction anything interaction based should be immediately category here i think the tensor network step fails because like a node is just a point but in category theory you can add space to that point by making the point something else and then you're working in this modulized space so that was my thought that about like spawn and you know when you were talking about spontaneous thoughts whoever had the question about that i mean i also think that like our work on the pro finite condition chris could also account for like thought whatever that is right because the thought doesn't really last right voice speech doesn't last the speech fades based on the composition of the molecules in the air but something remains i mean we all continue it seems to be that holographic copy of the pro finitely mini right in the diamonds and stuff so i just want so for syria you know i think a bunch of this category theory could really help you with uh you know uh you know it's like you have this object and topology called like a sheaf which is this very very complex structure but it attaches to a topological space and it's like a little bag and it keeps track of all the data you know and uh so if each space has these structures attached to it that keep track of what happens locally you can look at the modulized space and the isomorphism classes of those little bags so what's happening in the isomorphism space with the little bags probably seems to be what this like global consciousness things is so you can upgrade that one dimension using

category theory you can work in a stack and a stack is going to be a sheaf that doesn't take values in sets anymore all of quantum theory has taken place in sets and i think it needs to be a little bulldozed like it needs to go up if you can take values in categories then in each point you have so much more space to move around right so then if you actually have this object that takes values and categories and not in sets then you've allowed yourself more space so that you could have these phenomena like the the planera and stuff like that but if you're just working in like uh like points and like dots and like nodes connecting i think it's just not enough but you can do all these other but um i'm not sure if i answered anything but yeah chris i think that regener i was going to ask if you thought would like our which we need to pick back up the chromatic you know our chromatic type over uh temporal logic if that could actually work with this i'm fascinated by the regeneration and the stop mechanism which is weird you know yeah i'll just quote mike levin telling killing me over and over again and everyone else who will listen that the most important question of biology is when to stop yes now how how how does how does the embryo know that it's done now i can stop i'm an adult no one knows and that the question even isn't even asked that way but when you think about it it becomes it is a key question because it's it's related when you said like uh our experiential phenomenon uh capacity is just this you know the the tip of the non-experience so it's almost like if the stop is the iceberg of everything else i'd like to see what's the everything else you know thanks for these points it's really an amazing mapping that is rigorous and something that we should be learning more about how the point can be just the tip of the iceberg kind of where it's piercing something that's a lower dimensionality and so what do we do with that so i'm going to go to a question from chat and then can i ask can i ask a second part of the question i was working with go for it it's this is and feel free to just be like eh it's not great just just move you know not answer it if it but i i can't help but analogize um something that i just yeah i cannot ever let go which is um in electrical engineering or in electronics essentially you know you have this well in anything but i i really like electronics because it's very tangible you have the real component and the imaginary component of of of signal you know that goes across an interface and so i'm always really fascinated by that because it always seems like a really good analogy to this thing that we can look at through math or quantum mechanics but with electronics you actually have you know a thing that you can put on your desk um because you could think of this this this phase change that happens at interfaces as the context and and so i've always been interested in in trying to look at that thing that gets lost without looking at the at the current or the voltage like measuring at a point but actually just keeping

track of the context as things move across interfaces in a circuit it's not even a question i apologize for that but if it brings up any thoughts either of you two if you'd like to give a thought otherwise it stands and i love how many ideas people are bringing up this will definitely be an episode for people to re-listen to and make some knowledge network stuff uh all i'll say is phase is extremely important this is important it's a very it's like phase according to whom so chris and i really torn apart this reference frame idea like according to whom at what point if phase was like non-local then everything would be the same thing but so what point do you actually have you know electronics you assume that you have a free that you're like local and stuff but um electronics are fascinating it's still like kind of tangible but if you work on that that space that's pre-actualization i think it's very just like for those presented it's very very difficult to say who you are what the boundary actually is and if the boundary um like a boundary in mathematics is like an asymptote or something right so it's like a horizon or a koshi horizon or something like that so inside that very notion of a boundary you already have causality and then this you know tensor product and there's all kinds of other tensor part there's wedge products and you know it's it's i think it's really hard to say what a phase is without bringing into play all these other reference frames that are um so fragile cool i'm going to do a question from the chat and then i have a simple question so a question from the chat is do you think a reformulation of quantum mechanics is necessary in order to tackle the problem of consciousness or life from a full mathematical generality i'm mainly thinking of the notion of intrinsic information from tishbe's paper information theory of decisions and actions but also maybe other approaches like self-organized criticality more generally i'm curious how necessary is it to bridge our understanding of thermodynamics information theory and biology or control theory to have a good understanding how far along are we and are there any concrete problems for a mathematician i read what they say you know that's that's an omnibus question i i guess my my first answer would be yes uh we do need to bridge all those all those disciplines uh in order to make progress um whether it will tell us something deep about what experience is in some ontological sense i have my doubts about uh whether it will tell us something interesting about what experience um in a sense does how it works uh i think it may tell us something very important about that but yes i i i do think that these disciplines need to be bridged i i think we're a bit of an impulse that's been imposed on us by the disciplinary structure of science thanks for the answer i'm gonna ask my simple question and then we're gonna go to stephen and blue and also sarah i know you had some other kinds of questions that were a little bit less about the math but here's my simple question where does

come into play or for those who are just learning about active inference and curious to how it might be the bridge for all these different areas um what do you think active inference is doing here well i as i said at the very beginning i i think of active inference in terms of fundamental decision that that every system has to make all of the time between of do i explore do i act or do i retreat and and and try to do a kind of post-mortem on just what just happened and learned something so do i modify my priors or do i try to modify my my experience i think thinking of active inference in terms of what reference frames are being deployed it just allows us to pose that question a little bit differently it allows us to think of the question in the form of do i make another measurement do i act on the world or do i modify my reference frame right you can think of a reference frame uh like a category like a cognitive category so we have cognitive categories like this is a cat and that's a dog and this is a table and this is another person so modifying a reference frame can be thought of as going into your apparatus and changing the way it works you can also think of it as modifying my category modifying the things i'm capable of seeing and i i think you asked in the in the previous discussion last week what does the theory say about the the capabilities of organisms and or does it does it put a limit of what organisms can do and i think in a sense that's the question we're trying to answer to to what extent are you organisms have flexible capabilities to what extent can they retool their cognitive systems even if it's even if it's just e coli to what extent can can something like a microbial population retool what it's capable of and again biology gives us these beautiful examples in terms of things like lateral gene transfer where uh microbes in a community can pass around genes and so can can change very radically what they're capable of doing so i mean from an active inference point of view that's a question do i do i keep going with the capabilities that i've got or do i go shopping for some new dna cool it's like the active turn we don't know how far or what else what other stones are going to be turned when we actually start thinking and doing instead of just thinking about doing so we have stephen then blue then scott very excited to hear you talking about the work of my mike levin because his work's very uh when i saw the video of his work recently like last year was like wow it started to bring the whole idea of swarming and how swarming actually is underpinning a lot of what happens and the the the work with bioelectrics is really interesting and i i've been interested in how if that bioelectric aspect seems more maybe more closely tied to the sort of quantum transitions um and um just maybe a little clarity on a couple of things because often traditionally let's just ask the question and clarify it okay um so the question is um when you have these quantum effects you know normally they may be just you think of things flipping between one state and others

another state in more sort of basic chemistry thinking about like energy levels or something like that but you have many levels potentially so is it a case that you're able to still infer from the noise what's going on with quantum states and therefore this gives us a way to not have to just go for like the attractor theory and complexity or catastrophe theory or phase theory that we use at the moment but you're able to still extract quantum state information in ways that's not just classical like one energy level to another energy level but you get this more higher information context yeah good question i i think what you're really asking is uh where are the experimental tools that let us uh probe this domain and um i wish i knew and this is something that mike and i talk about for example how could we go about looking for contextuality effects for example in things like bioelectric effects and worm regeneration and i don't know i think i think there's a huge opportunity space here for for people to think about uh ways of doing experiments that we can't experiments that we can't do now or have never thought about being able to do probe a domain of correlation that is non-classical or super determined or however however you want to think about it and i think that the the more we can think of biological systems in in communication theoretic terms the better prepared will be to design experiments to look for what are really communication theoretic effects and quantum theory is really thank you cool blue and then scott so first because shanna dropped off i don't know if she's gone or if she's gonna rejoin and it's kind of a follow-up to some of the stuff that she was saying but i can still ask it um but let me pass to scott and then i'll go we'll go to scott and hopefully um we can have some future category streams to really learn more about this area so yeah scott and then back split great um thank you this is my smile is hurting my face it's so such a great discussion um the two things i want to talk about embodiment and this is related and one is social level experiments so the first one is embodiment it seems like we have these internal states or embodiments of prior external states or their learned and so it's kind of interesting because the external states in the internal states have bearish significant relationship through time not just through that individual organism's markov blanket but by the inherited and other factors that are become the priors right and so it's kind of interesting because they have embodiment externally and internally maybe that's extremely fluid right because if you talk about systems at different levels they're different types of systems cell systems social systems organisms organizations they all have different affordances and their embodiments are not necessarily they can be um they're not necessarily inconsistent when they're different right you can have an embodiment of something uh if i eat a lot of fatty food i become fat but that doesn't you know anyway that's one thing is the embodiment if you talk a little bit about that

for me from my perspective if you're looking at risk and change when you information being shifted around as a lawyer ex-lawyer i look at risk and risk being moved around and so one of the things that seems to me when you have the embodiment question is that among the relationships then you get the question since entropy or disorder is going to increase universally what we're doing is we're um managing local meg entropy in these systems right we're sharing uh local entropy and so the question is is that dynamic something that um tends to dissipate the markov blanket as a standalone thing maybe markov blankets are a general thing that's one question okay let's get that let's try to do one question because it's really you know in a body domain let's just try to yeah thanks for the really great question go ahead yes okay um this is a wonderful question i'll make uh two comments one is in terms of of reference frames which i always come back to um i think what what what quantum theory contributed to the to the notion of a reference frame is it insisted that we regard reference frames as embodied as physically implemented and stop thinking of them as merely abstractions and we can we can go back for example to the idea of cognitive categories you know my my idea of a table is not an abstraction it's a it's something that's implemented in my brain and it can be changed by operating on my brain and i can change it that way we don't know how that works right but but what the what the the embodied movement or the the 4e framework whatever what that forces cognitive science to think to think about is these are not abstractions we're talking about things that have been implemented so i think that's an incredibly important point um the second point i want to make is boundaries and i showed that picture of uh right alice looking at her at her holographic screen and listed the things that she doesn't know but i could have added to that that alice doesn't know where her boundary is if if i'm a system and i'm i'm looking out at the world through my markov blanket or through my my encoded screen then what i'm seeing is this array of information that's painted on my screen right this array of bits and and fristen's point and the point of the holographic principle is i don't see what's on the other side but you can flip that around it's always symmetrical i don't see what's on my side either and that's that's why i tried to make this point very briefly about memory how if we think of memory as then it's got to be written on the screen because that's the only focus of any classical information so if i'm looking at my screen the thing that i don't see is myself and since i don't see myself and i don't see the other side i just see the screen i have no idea where it is so when you when you raise this question a social system or or any other kind of complex system um we as as observers of that system are using our reference frames or our cognitive categories or our however you want to describe them to

draw a boundary around that thing that boundary is our imposition it's not the system's imposition the system doesn't know where its boundary is there a talk is there a tautology in there though because that's interesting right because we're using our system we're drawing the system within our boundary it seems like that's an intrinsic quality of the thing that system can can a system know it's open kind of system know others and you can only know it through the boundary but you so you can't know yourself because you're not written on the boundary so it's kind of it well let me let me get to the next question because it relates to that so the next question but the quick answer to that question is yes yeah i like that's a good question it's good when there's a binary answer so the next question was i was wondering if we can um if the for experimentation this is the question of answering experimentation if and to the extent that this is there's scale independence and active inference and so that it applies at cell levels and social etc why can't we look at the social level to run experiments and then apply those what we come out with at other levels so it's similar to when you do a qubit at a macro you know they do qubit qubits that are manifested in different ways in different kinds of systems you know ionic qubits and things like that so you can measure them in different ways so we could be observers of the system at the social level to identify some qualities and the reason i'm asserting that as a lawyer even though the social has a lot of variables what we do is as lawyers is create fewer variables right we take a meaning thing we say hey everyone in this area of the statute has to behave this way and so then internally everyone can absorb and embody the idea that i know everyone's going to stop at a red light i know i'm going to stop at a red light that's the story of the red light red means stop you're taught it as a kid so it becomes instantiated and kind of axiomatic so one of the things about the law is that i'm going at it from a slightly different way of effectively going in and trying to non-variables in the markov blanket that are shared right and so and then so i wonder about the embodiment relates in a way i think to the experimentation question because can we do let's call them artificial embodiments into the internal states through things like law or folkways or shared meaning or norms or things like that that we can then experiment against is what i'm wanting cross-cultural experiments things like that that then may have application at other levels i i think that's i think that's a fascinating thing to try to follow up on i would say one of my one of the things that seems most mysterious to me is language is is the very possibility of classical communication of what what you're talking about a physicist would call super determinism and super determinism is essentially the same thing as if i don't have a choice about what reference frames to deploy if there's a correlation in advance super determined uh by the past

between the reference frames i use and the reference frames you use then and in an important sense we're not independent systems anymore right we're we're entangled but that's what language is in in some sense and so i i i agree with you we should be able redescribe communication generally in these terms of of reference frame sharing reference rank choice of super determinism and apply some of this math that's been worked out in other places to reinterpret communication and i i'll mention a paper by a guy named alexi grinbaum it's a philosopher of science at saclay and it's a paper on uh device independent methods in quantum but the very last i see sarah you're nodding your head you probably read this paper one of the very last sentences in the paper is he says this redefines what physics is about physics is about languages and i think that's a phenomenally profound statement i think he's right nice let's continue in order so we got blue and then stephen and then sarah so shanna's not here i think she said in the chat that she um had to run but um i want to touch back on what she said like when a thought passes that there's still something there and also chris to what you were saying earlier about memory and what uh sarah was saying of keeping track of what gets lost and so something that i gleaned from this paper now i have like no background in category theory at all um as you may have been able to tell from the introduction video right so um the so it seems like that there's this hierarchical nature of some mathematical summation that is like able to occur like via these methods and i just was wondering if you're familiar with eric cole's work um he put out a paper last year on causal geometry and um like he's done some coarse graining stuff and so so is there some relationship between like that kind of coarse graining or like dimension reduction and this category theory work or is the category somehow able to encompass um the entirety of the mathematical problem in a better way or a different way than say core screening um yeah good good question um uh yes i i do know eric's work and i talk to him almost every week because he's also part of mike levin's lab and so we have discussed this question of course graining and causal emergence and effective information all these concepts that that eric has been working with um i think are very closely related to the concepts that jim and i have been and we're we're coming at them from such different starting points that it's not clear at least to me exactly how they fit together but what a what a coconut diagram does is coarse-grain it it represents a an abstraction of a set of logical relationships or or it represents a an abstractions of a set of criteria into a coarse-grained larger scale if you will uh criterion that covers all of the cases below it but in which information is both lost and gained i i think the the the best example at least for me for thinking about it is again going back to cognitive categories if you think about you know your category for a table or another person or whatever

you you lose information in that abstraction process but you also gain information by tying all of those exemplars together and each exemplar is then in some sense encoding through the rest of the network of its consistency in some sense with other exemplars that may have different uh detailed characteristics so at the token level they look inconsistent but at the tight level they're consistent and the um the coco diagram represents that that very complex relationship so when one is doing a course grading and it works the the information in the course grain state is capturing just this kind of abstract information it's the it's the abstract information that's actually useful and that captures what is consistent between the the fine-grained exemplars and by organizing them that way it allows each of them to represent the fact of its consistency with this larger set thanks i also like i'm curious to see because i do think that it's gonna like you guys are we'll lock together eventually at some point um you know maybe in 10 years but but i'm curious to see how that um you know kind of dynamic interplay unfolds will also lie also um mike living will be on active inference live stream on july 13th so we can talk before then or after then all right awesome so we have um a little bit less than 20 minutes left so we'll just do you know questions and try to get a few more questions in from the chat as well stephen and then sarah thank you yeah deep breath there's a lot a lot to go through here um i was gonna sort of build on that coarse grain question and maybe just stepping back at some of the bigger implications of contextuality um and how that is thought about so um you know the principles that you talked about or the paradigms um so i'm curious about how you see this maybe impacting contextuality um and the way people think about that let's address that and then continue stephen let's just get one question and one answer okay okay i guess i didn't i didn't actually hear that as a question sorry guys okay thank you sorry yeah i just i just finished that thought but basically just in terms of contextuality and coarse graining and you know how much granularity is thought about um and how that has a bigger that that general idea of how to think about contextuality um i'd be interested in your thought about that as how that sort of has a bigger set of legs and one of the reasons i say that is because when people often talk about quantum in a sort of broader world out there they often think about this transpersonal it's all the same and it's unifying and it's actually decontextualizing in some ways you know it's almost some ways this is quite revolutionary to me to really see quantum effects as being contextual you know being in that way so you know that maybe ties in with david bowman's work on implicit order and the awareness of things so rather than always being about the whatness and going up indigenous approaches are very much about where you are and how you are what's your embeddedness in your niche in some ways it's like quantitative

now when i first heard about your paper i was thinking oh this is you hear quantum you think abstract but actually it's doing something very different i just wondered how that might have impact in other practices okay so so i i heard a number of different ideas here that that are are parts of questions one is to do with with course training and contextuality and you put your finger very precisely on the danger of course grading uh and this this you can also state this in terms of the frame problem it's it's often the little details that come back to bite you and when you coarse grain you're specifically getting rid of those little details so this is uh goodness it it it's like the the metaphor of the hammer and the nail right if any choice of reference frame is effectively a choice of course grading and choosing a reference frame choosing a course greeting determines what you can see but that also determines what you can't see and it's what you can't see that is responsible for the contextual that you can't avoid by by by you know um learning more about the context if you've if you've arranged your measurement procedures in a way that prevents you from seeing them then you can never take them into account when you're building your probability distribution so so coarse graining is something that we're stuck with but it's it's very much a double-edged sword right it allows us to get around in the world it allows us to see these similarities it allows us to learn things but it forces us to ignore things that may turn out to be important or may turn out to have been thanks for the really interesting answer there um so so with respect to you to your other question about if you will embeddedness uh yes i think that's crucially um let's go to sarah i'm just going to say yeah that that that whole embeddedness is huge for the inactive ecological cognition work and sort of the the pragmatic turn in psychology so there's a lot of implications of that so i just think that's very exciting so thanks yeah yeah i i'm very attracted to this sort notion that ecosystems are are in a sense extended organisms etc that whole way of thinking in biology is i think very important and also to connect that it's like there's this embedded practice and then we're talking about the physical electronic circuits so it's almost like across domains we're taking an action-oriented and a realist manifested focus but then seeing the what is actually manifest as the tip of the iceberg in another sense but it's totally real for what it really is which is our measurement and then also it's the tip of an iceberg because we're also in the system and those kinds of systems don't just arise out of nowhere so sarah and then scott in our last couple of minutes but thanks everyone for this really awesome discussion maybe sarah yeah good god yeah um so am i to understand that we have another one of these because i i would love to ask chris you know these other questions but obviously not now there's just not enough time we will be discussing the paper next week but um we did not plan without totally no pressure but

we're doing the same time next week but no expectation okay no prior okay um yeah in some ways this is like a media um just rephrasing what's already been said but it's actually right up right up your alley steven um i was actually just in writing in my paper you know about well what methods you choose depend on what met what questions you ask um related to reductionism related to this whole theme um but i i wonder this is like super abstract um but like i wonder about the the role of play because i was thinking about you know like i was thinking about um there's so many situations in science where you're not you know like when you look at the paper you're like oh yeah they're doing science but in reality they're just poking at right like so it's just this way of being with the world where you're like oh i wonder what happens if i you know but they're not really reducing they're just it's like so it's like play and so i've really wondered about about how to bring that to the foreground about what that what that is in science you know and so i again i apologize but you're you always have a way of coming up with a comment so hopefully you'll have a thought on this oh yeah if it's not fun you're probably doing it i was just thinking of like an antidote to reductionism like play seems like a way that you're not taking things apart you're just kind of you're with it in some way well i to take this back to active inference and to to think of it uh almost in in robotics terms right how what the people in robotic and developmental robotics are are trying to do is understand what motivates a system engage with the world with this kind of innocence that i think you're referring to to just explore and and see what's learnable right the uh one of the one of the key questions is is um and pierre etier is is the one who i think in the literature has emphasized this more than anyone else is how do how do you learn what parts of the world are learnable with the resources that you have so that you can avoid spending all of your resources trying to learn things that you're not capable of learning so how do you how do you how do you focus your exploratory behavior in the parts of the world where it can actually be productive this is sort of a meta question on top of active inference and it has to do with bayesian precision and things like that but it's your expectations about learnability and then there's a flip side of this in evolutionary theory right the whole idea that that what organisms are doing in evolution is trying to improve their evolvability uh so they're they're they're trying to increase if you will the flexibility of their lineage to cope with with whatever the environment can throw at them and this is very much i think what what's happening in play as you put it you're trying to to engage with the world in a way that is wanting to know what the possibilities for further engagement are thanks for the answer and speaking to play it's almost like you start with what you have the rubber band the tin foil so you start with the actual but then you you think about

what if and you think about adjacencies even categorical adjacencies um so that was a pretty interesting so and then also a comment on what you said about the expectations of in active inference lab we would hope to learn by doing and applying and update people's expectations to increase their um understanding of they can learn active inference basically we'll figure out other ways to say it more directly but will manifest through observations on people's screens that they can learn active inference and then it's almost like in the cognitive dimension our collective and our individual cognitive evolution will undertake another inflection point when we believe that we can learn so that will be really powerful so scott yeah along those lines just in terms of further leverage i want to talk a little bit about the buy or build decision that every organism makes and the idea is do you do you develop it internally or do you get it outside right and so it seems like one of the things that struck me is we've talked about the being tied to language one of the things i've been fussing with lately and i've this is the question i've been fussing with the idea that the mind resides in language and the brain is just an antenna tuned to the mind now the antenna would be the markov blanket right or the the action of the markov blanket is a continuous dynamic tuning and the so that you're if you're born into a cultural context in my place very quickly my sister adopted a five-month-old chinese baby she was raised in central pennsylvania she looks chinese she doesn't speak chinese doesn't like chinese food she's not chinese right so her mind is a creature of her environment in body so one of the things in terms of affordances you know we talk about resources internally to develop the markov like it you know you can really shortcut that if you say hey i know that my resource internally is i should time myself to to daniel because he's awesome and he knows all the stuff i don't have to know it i just have to know if i have a question i'll go ask him right so you start to have something like it's like eukaryotic revolution and multicellularity you start to have this de-risking and leverage at larger scales made possible by the fact of synthesis of multiple markov blankets they can be synthesized around common ideas or common problems so the the question is in terms of that separateness we see ourselves and our markov blankets and our minds as separate but it seems like maybe they're not maybe maybe the mind is all one thing and we have these instantiations of it that we call different there are different iterations of that markov language can you comment on that kind of macro and micro level uh notion briefly um i'll try i think this is all good thinking i i view this in the language of in a sense extended organism thinking and evolutionary theory which is very closely connected to to multicellularity and evolutionary uh which is closely connected to the sorts of of faculty multicellularity you see in microbial communities um i think from the from i think we see

examples of this throughout biology and um so the question becomes how do we take this into the human sphere or as i think you posed it very clearly uh how do we realize that it's already been taken into the human sphere we just don't quite get it yet and and i suspect that you're right uh that we just don't quite get it yet and you know i i remember in graduate school someone said you're what you're learning is how to read the literature yes you don't need to learn any facts and that has to do with security right and in new forms of threat and integrity and privacy issues all these new allergic problems are not necessarily new problems okay and that's why i wondered about the biology and the biomimicry you see that we can how can we actively pursue biomimicry thank you cool sorry you're a little quiet at the end there scott but um with this um close the hour um chris really thanks so much and also um shauna for joining with your expertise but really this was a special event for us we would always welcome the opportunity to speak with you or colleagues to learn more and any other final thoughts um that you'd like to give in closing chris uh yeah feel free to send me an email my my address is on my website chrisfieldresearch.com all the papers i published recently are there um but yeah i'd love to continue discussing awesome well thanks a lot and um next week we're going to be having a follow-up discussion to really digest this definitely re-listen to it and go from there so thanks everyone for joining and we'll see you another time